

Report: Laptop Price Prediction Model Development & Evaluation

1. Introduction

About Dataset _ What Factors Affect Laptop Computer Prices?

Several different factors can affect laptop computer prices. These factors include the brand of computer and the number of options and add-ons included in the computer package. In addition, the amount of memory and the speed of the processor can also affect pricing. Though less common, some consumers spend additional money to purchase a computer based on the overall “look” and design of the system.

In many cases, name brand computers are more expensive than generic versions. This price increase often has more to do with name recognition than any actual superiority of the product. One major difference between name brand and generic systems is that in most cases, name brand computers offer better warranties than generic versions. Having the option of returning a computer that is malfunctioning is often enough of an incentive to encourage many consumers to spend more money.

Functionality is an important factor in determining laptop computer prices. A computer with more memory often performs better for a longer time than a computer with less memory. In addition, hard drive space is also crucial, and the size of the hard drive usually affects pricing. Many consumers may also look for digital video drivers and other types of recording devices that may affect the laptop computer prices.

Most computers come with some software pre-installed. In most cases, the more software that is installed on a computer, the more expensive it is. This is especially true if the installed programs are from well-established and recognizable software publishers. Those considering purchasing a new laptop computer should be aware that many of the pre-installed programs may be trial versions only and will expire within a certain time. To keep the programs, a code will need to be purchased, and then a permanent version of the software can be downloaded.

Many consumers who are purchasing a new computer are buying an entire package. In addition to the computer itself, these systems typically include a monitor, keyboard, and mouse. Some packages may even include a printer or digital camera. The number of extras included in a computer package usually affects laptop computer prices.

Some industry leaders in computer manufacturing make it a selling point to offer computers in sleek styling and in a variety of colors. They may also offer unusual or contemporary system design. Though this is less important to many consumers, for those who do value “looks,” this type of system may be well worth the extra cost.

Parameters

The parameters used in the dataset are:

1. Manufacturer - The company that manufactured the laptop
2. Category - The category to which the laptop belongs: This parameter is mapped to numerical values in the following way:

Category	Assigned Value
Gaming	1
Netbook	2
Notebook	3
Ultrabook	4
Workstation	5

3. GPU

The manufacturer of the GPU. This parameter is mapped to numerical values in the following way:

GPU	Assigned Value
AMD	1
Intel	2
NVidia	3

4. OS

The operating system type (Windows or Linux): This parameter is mapped to numerical values in the following way:

OS	Assigned Value
Windows	1
Linux	2

5. CPU_core

The type of processor used in the laptop: This parameter is mapped to numerical values in the following way:

CPU_core	Assigned Value
Intel Pentium i3	3
Intel Pentium i5	5
Intel Pentium i7	7

6. Screen_Size_cm - The size of the laptop screen is recorded in cm.

7. CPU_frequency - The frequency at which the CPU operates, in GHz.

8. RAM_GB - The size of the RAM of the system in GB.
9. Storage_GB_SSD - The size of the SSD storage in GB is installed in the laptop.
10. Weight_kg - The weight of the laptop is in kgs.
11. Price - The price of the laptop is in USD.

Objectives:

The objective of this study was to develop and evaluate multiple machine learning models to predict laptop **Price** based on selected hardware features:

- RAM_GB
- CPU_core
- CPU_frequency
- Storage_GB_SSD
- GPU

The workflow followed a complete data science pipeline including data preprocessing, exploratory data analysis (EDA), feature selection, model development, and evaluation using multiple regression and ensemble techniques.

2. Data Preprocessing & EDA

2.1 Data Cleaning

- Checked for missing values and inconsistencies
- Handled duplicates and corrected data types
- Standardized numerical formats

2.2 Exploratory Data Analysis (EDA)

EDA was conducted on both **raw and encoded datasets**.

Variables analyzed:

- Categorical: Manufacturer, OS, Category
- Numerical: Screen_Size, Weight, RAM_GB, CPU_core, CPU_frequency, Storage_GB_SSD, GPU, Price

Key steps:

- Distribution plots for numerical features (Price, RAM, Storage, etc.)
- Boxplots for outlier detection

- Correlation heatmaps for feature relationships
- Bar charts for categorical distributions (Manufacturer, OS, Category)
- Comparison plots before and after encoding

Key Insight from EDA:

- **RAM_GB and Storage_GB_SSD showed strong positive correlation with Price**
- CPU features contributed moderately
- GPU presence had a nonlinear impact on Price
- Categorical features improved interpretability but were not included in final regression models

3. Feature Selection

Final selected features used in modeling:

Feature	Reason
RAM_GB	Strong correlation with price
CPU_core	Performance indicator
CPU_frequency	Processing speed influence
Storage_GB_SSD	Major cost driver
GPU	Performance & gaming capability

4. Model Development & Evaluation

4.1 Model Performance Summary Table

Model	Features Used	R ² Score	MSE	CV R ² (if available)
Simple Linear Regression	RAM_GB, CPU_core, CPU_frequency	0.5468	13.93e12	-
Multiple Linear Regression (3 features)	RAM_GB, CPU_core, CPU_frequency	0.5688	13.25e12	-

Model	Features Used	R ² Score	MSE	CV R ² (if available)
Multiple Linear Regression (5 features)	RAM_GB, CPU_core, CPU_frequency, Storage_GB_SSD, GPU	0.5725	13.14e12	0.5886
Polynomial Regression (RAM only)	RAM_GB	0.5733	16.47e12	-
Polynomial Regression (Pipeline Multi-feature)	All 5 features	0.6539	13.36e12	-
Ridge Regression (Best $\alpha=1$)	All features	-	-	0.5886
Random Forest (CV Mean)	All features	-	-	0.6458
Tuned Random Forest	All features	-	-	0.6605
Gradient Boosting	All features	-	-	0.6643
Tuned Gradient Boosting	All features	-	-	0.6747
XGBoost	All features	-	-	0.6325
Tuned XGBoost	All features	-	-	0.6700

4.2 Train vs Test Validation (Key Check)

Metric Value

Train R² 0.5652

Test R² 0.4105

→ Indicates **overfitting** in simpler model split evaluation

4.3 Cross Validation Summary

Model	Mean R ²	Std Dev
4-Fold CV (baseline model)	0.5611	0.0242

Model	Mean R ²	Std Dev
Multi-feature CV model	0.5886	-
Random Forest	0.6458	0.0590
Gradient Boosting	0.6643	0.0714
XGBoost	0.6325	0.0477

5. Model Comparison & Interpretation

5.1 Key Observations

Linear Models:

- Performance is moderate ($R^2 \approx 0.55\text{--}0.57$)
- Limited ability to capture nonlinear relationships

Polynomial Regression:

- Improved performance significantly when using multiple features ($R^2 = 0.6539$)
- However, single-feature polynomial model is unstable (higher MSE)

Ridge Regression:

- Helps regularize linear model but does not outperform tree-based methods

Ensemble Models:

- Consistently outperform linear and polynomial models
- Capture nonlinear relationships effectively

5.2 Best Performing Models (Ranked)

Rank	Model	CV R ²
1	 Tuned Gradient Boosting	0.6747
2	 Tuned XGBoost	0.6700
3	 Tuned Random Forest	0.6605
4	Gradient Boosting (baseline)	0.6643
5	Random Forest (baseline)	0.6458

6. Conclusion

Based on comprehensive evaluation across **linear regression, polynomial regression, ridge regression, and advanced ensemble models**, the following conclusions can be drawn:

1. Linear regression models provide a baseline understanding but are insufficient for capturing complex relationships between hardware specifications and laptop price.
2. Polynomial regression improves predictive ability, especially when multiple features are included, but still suffers from instability and overfitting tendencies.
3. Regularized linear models (Ridge Regression) improve generalization slightly but do not outperform nonlinear models.
4. Ensemble learning methods (Random Forest, Gradient Boosting, XGBoost) consistently achieve the best performance due to their ability to model nonlinear interactions between features.

Final Best Model Recommendation

✓ Tuned Gradient Boosting Regressor

- **Best CV R^2 Score: 0.6747**
- Strong generalization performance
- Robust against overfitting
- Captures complex nonlinear relationships effectively

7. Final Statement

Considering all evaluation metrics (R^2 score, MSE, cross-validation stability, and generalization performance), the **Tuned Gradient Boosting model** is the most suitable and reliable model for predicting laptop prices based on hardware specifications.

It achieves the best balance between **accuracy and robustness**, making it the recommended final model for deployment.
